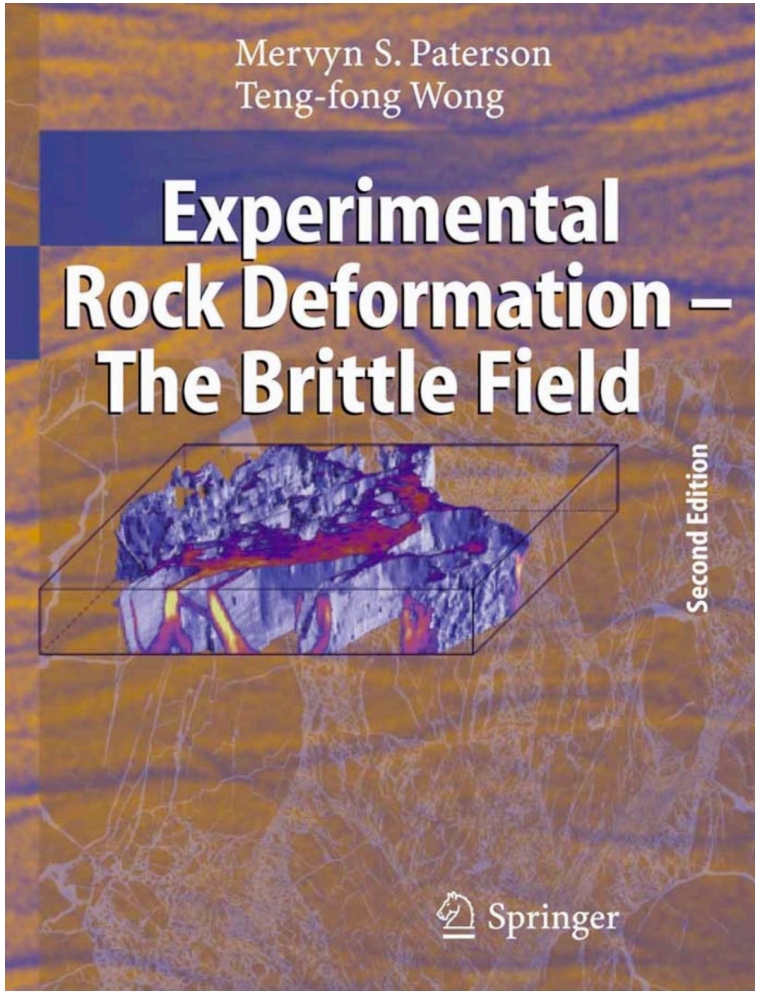
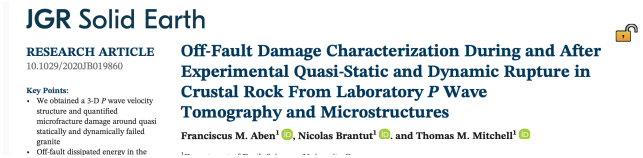
COURSE MAP

Where and How to study

Text book



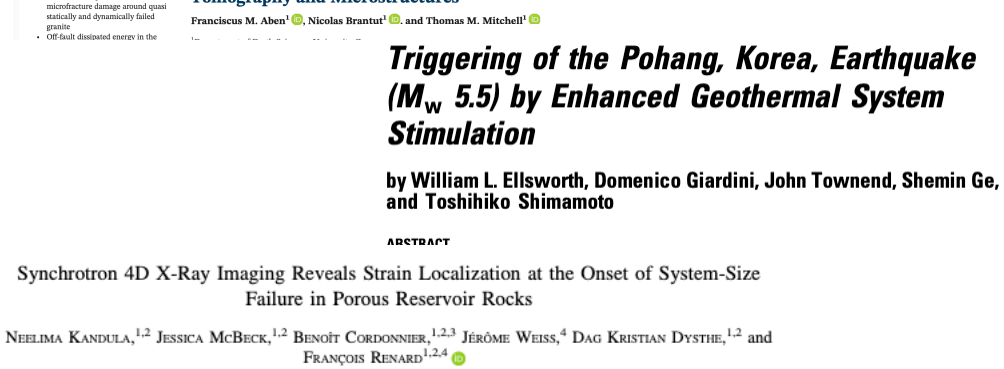
Scientific papers


JGR Solid Earth

RESEARCH ARTICLE
10.1029/2020JB019860

Off-Fault Damage Characterization During and After Experimental Quasi-Static and Dynamic Rupture in Crustal Rock From Laboratory *P* Wave Tomography and Microstructures

Franciscus M. Aben[1], Nicolas Brantut[1], and Thomas M. Mitchell[1]


Triggering of the Pohang, Korea, Earthquake (M_W 5.5) by Enhanced Geothermal System Stimulation

by William L. Ellsworth, Domenico Giardini, John Townend, Shemin Ge, and Toshihiko Shimamoto

ABSTRACT

Synchrotron 4D X-Ray Imaging Reveals Strain Localization at the Onset of System-Size Failure in Porous Reservoir Rocks

Neelima Kandula,[1,2] Jessica McBeck,[1,2] Benoît Cordonnier,[1,2,3] Jérôme Weiss,[4] Dag Kristian Dysthe,[1,2] and François Renard[1,2,4]


Journal of Geophysical Research: Solid Earth

RESEARCH ARTICLE
10.1002/2017JB014964

Critical Evolution of Damage Toward System-Size Failure in Crystalline Rock

François Renard[1,2], Jérôme Weiss[2], Joachim Mathiesen[3], Yehuda Ben-Zion[4], Neelima Kandula[1], and Benoît Cordonnier[1,3]

Fracture Network Localization Preceding Catastrophic Failure in Triaxial Compression Experiments on Rocks

te.pdf

ssica McBeck[1]*, Yehuda Ben-Zion[2] and François Renard[1,3]

[1]The Njord Centre, Departments of Geosciences and Physics, University of Oslo, Oslo, Norway, [2]Department of Earth Sciences and Southern California Earthquake Center, University of Southern California, Los Angeles, CA, United States, [3]University Grenoble Alpes, University Savoie Mont Blanc, CNRS, IRD, ISTerre, Grenoble, France

Each resource has a different job

LECTURE SLIDES	Follow the concepts, figures and questions used in class.
PATERSON & WONG	Use assigned sections to deepen experimental mechanics.
PAPERS	Read how a research claim is supported by figures.
DATA + NOTEBOOKS	Reproduce calculations and test your own interpretations.
INTERACTIVE GUIDES	Practise a model, then explain what the model leaves out.

A repeatable routine around each lecture … Ideally

BEFORE	20 min	Prepare a question about the previous lecture.
DURING	Class	Annotate decisions, measurements and surprising results.
AFTER	23.5 hours	Recreate one graph or calculation. Write a claim, its evidence and one limitation.

Use AI to test explanations, then check against evidence

EXPLAIN	“Explain why porosity and permeability differ. State assumptions.”
QUESTION	“Ask me three questions on ultrasonic velocity. Wait for my answers.”
CHALLENGE	“Offer two other explanations for this trend and tests to distinguish them.”

If you only look for answers you are just hurting yourself

A plausible answer can overstate what a signal means

“The fall in ultrasonic velocity proves that a continuous fracture has formed across the sample.”

What was observed? What does “proves” assume? What other measurement would you seek?

Treat this as a sample AI claim to audit, not an experimental result.

You remain responsible for every figure and claim

01 Check equations, units and citations against source material.

02 Keep raw data and a reproducible analysis.

03 Disclose the tool, its role and what you checked.

Example note: “I used AI to clarify terminology; I checked the definitions and analysed Figures 2 and 4 myself.”

EXAM · Group PROJECT

Choose a paper and build an evidence-based argument

Select one paper from the approved list.

01 Explain the research question and experimental design.

02 Interpret two central figures in detail.

03 Connect the evidence to at least two course modules.

04 Judge what the data support and what remains uncertain.

Written analysis followed by an individual discussion

4–6 pages

Submitted before the exam

15 + 15 minutes

Presentation + questions

Figures and evidence	30%
Connections to the course	25%
Experiment and measurements	20%
Critical evaluation	15%
Presentation and discussion	10%

TAKEAWAY

The course asks you to defend what the data mean

What did we measure?

How was it transformed?

How strong is the interpretation?

Exit question: Which piece of evidence would you seek first in the Pohang case?